

MATERIAL SAFETY DATA SHEET

PROFENOFOS 40% + CYPERMETHRIN 4% EC

1 SUBSTANCE/PREPARATION AND COMPANY IDENTIFICATION

Substance Name	Profenofos 40% + Cypermethrin 4% EC
Chemical Family	Organophosphate Insecticide/acaricide/Synthetic pyrethroid
Use	Agriculture Use
Company Name and Address	Coromandel International Ltd, 1-2-10, S.P. Road, Secunderabad - 500 003 Telangana, India Tel: +91 40 27842034/7212
Emergency Information:	
Emergency contact address:	Coromandel International Ltd, 1-2-10, S.P. Road, Secunderabad - 500 003 Telangana, India
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2 COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	Profenofos	O-(4-bromo-2-chlorophenyl) O-ethyl S-propyl phosphorothioate
	Cypermethrin	(RS)- α -cyano-3-phenoxy benzyl (1RS, 3RS, 1RS, 3SR)-3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropane carboxylate
CAS No	Profenofos	[41198-08-7]
	Cypermethrin	[52315-07-8]

Chemical Composition

Profenofos a.i	40.00% w/v
Cypermethrin a.i	4.00% w/v
Other ingredients	56.00% w/v
Total	100.00% w/v

3 HAZARD IDENTIFICATION

KEEP OUT OF REACH OF CHILDREN - CAUTION

Harmful in contact with skin, if swallowed and by inhalation. May cause sensitization by skin contact. Avoid contact with skin. Avoid breathing of spray mist. Avoid contamination of feed and food stuff. Keep out of reach of children.

Dangerous for the environment. Very toxic to aquatic organism, may cause longterm adverse effect in the aquatic environment. Toxic to Birds. Very toxic to honey bees.

4 FIRST AID MEASURES

If swallowed	Call a physician immediately for treatment. If medical assistance is not readily available, and poisoning symptoms are severe, vomiting should be induced by tickling the finger on the back of the patient's throat. Do not give anything by mouth to an unconscious person.
If eye contact	Hold eyes open and rinse slowly and gently with water for 15-20 minutes and Call a physician
If skin contact	Take off contaminated clothing. Rinse skin with plenty of water for 15-20 minutes. Call a physician if necessary.
If inhaled	Move the person to fresh air. If person is not breathing give artificial respiration and call physician immediately for further treatment
Note to physician	Inject 2 to 4 mg atropine sulphate i.v or i.m every 5 to 10 minutes until signs of atropinisation occur. The patient must remain atropinised for a sufficient length of time (24 to 48 hours) to prevent subsequent poisoning due to delayed resorption.

5 FIRE - FIGHTING MEASURES

Flash Point	> 35.5° C
Extinguishing media	Dry chemical, Carbon dioxide, foam, water spray or fog
Hazardous combustion products	Combustion products are toxic/and or irritant. Haloid acids.
Special Fire Fighting procedures	Wear Self contained breathing apparatus with full protective clothing. Fight fire from upwind and keep all non-essential personnel out of area.
Unusual Fire and Explosion Hazards	If water is used to fight fire or cool containers, contain runoff by diking to prevent contamination of water supplies. Containers in fire may burst or explode from excessive heat. Stay away from the area.

6 ACCIDENTAL RELEASE MEASURES

STEPS TO BE TAKEN IF MATERIAL IS RELEASED OR SPILLED:

For small spills, absorb with an absorbent material such as pet litter. Sweep up and transfer to containers for possible land application according to label use or for proper disposal. Check for local authorities for proper disposal. Flush the area with water to remove any residue.

CAUTION: KEEP SPILLS AND CLEANING RUNOFF OUT OF MUNICIPAL SEWERS AND WATER RESERVOIRS

7 HANDLING AND STORAGE

Handling	Wash hands before eating, drinking, chewing, using tobacco or using the toilet. Remove clothing immediately if pesticide gets inside. Then wash thoroughly and put on clean clothing. Remove PPE after handling this product.
Storage	Keep out of reach of children and keep away from food, drink and animal feed. Keep container tightly closed when not in use. Do not contaminate water, food, or feed by storage or disposal.
Personal Protective equipment	Applicators and other handlers must wear: long sleeved shirt and long pants, chemical-resistant gloves. Barrier laminate, Nitrile rubber and shoes. Follow manufacturer instruction to cleaning and maintaining PPE.

8 EXPOSURE CONTROLS AND PERSONAL PROTECTION

Engineering Controls	Mechanical ventilation should be used when handling this product in closed spaces
Respiratory Protection	Incase of heavy exposure, wear: NIOSH approved organic vapor respirator in absence of mechanical ventilation
Skin Protection	Wear Protective clothing: long sleeved shirts and pants, hat, shoes and chemical resistant PVC or nitrile gloves
Eye Protection	Wear safety goggles with splash proof non perforated side shield or goggles
Other	Wear PVC or chemical resistant disposable overalls and PVC boots

9 PHYSICAL AND CHEMICAL PROPERTIES

Physical State and odor	Pale yellow color liquid with garlic odor
Specific gravity	1.08 g/ml at 31°C
Flash Point	>24.5°C (Able method)
Boiling Point	NA
Solubility in water	Emulsifies in water
pH	4.5-6.0 (1% solution)
Explosive Property	Non-explosive
Oxidizing property	None
Auto ignition Point	None

10 STABILITY AND REACTIVITY

Stability	Stable under standard conditions
Incompatibility	Incompatible with strong oxidizers and bases.
Conditions to avoid	Excessive heat and ignition sources
Hazardous decomposition products	Combustion products are toxic/and or irritant. Haloid acids.
Hazardous Polymerization	Will not occur

11 TOXICOLOGICAL INFORMATION

Acute Oral LD₅₀ (rat)	>1000 mg/kg	Acute Dermal LD₅₀ (rabbit)	>2000 g/kg
Eye Irritation (rabbit)	Mild Irritant	Skin Irritation	Mild Irritant
Inhalation LC₅₀(rat)	Not determined	Skin Sensitization	Non-sensitizer
Mutagenicity	Non-mutagen	Reproductive toxicity	None
Toxicity class	WHO (a.i) II EPA (Formulation) II	ADI (JMPR)	NA

12 ECOLOGICAL INFORMATION

Acute Oral LD₅₀ (Birds)	> 1000 ppm (Japanese Quail) (Technical)	Dietary LC₅₀ (bobwhite quail)	70-200 mg/kg (Technical)
Daphnia EC₅₀(48h)	1.06 µg/l (Technical)	Bees (Contact)	0.102 µg/bee
Acute LC₅₀(Fish) 96 h		Algae EC₅₀	1.16 mg/l (72 h) (Technical)
Blue gill sunfish	0.3 mg/l (Technical)		
Rainbow Trout	0.8 mg/l (Technical)		
Degradation	Rapidly taken up by plants and metabolized. Degradation to polar metabolites.		

13 DISPOSAL CONSIDERATIONS

Do not reuse product containers. Dispose of in sanitary landfill, or incineration or if allowed by state and local authorities, by burning. If burned stay away out of smoke. Pesticide wastes are toxic. Improper disposal of excess pesticides, spray mixture or rinsate is a violation of law. Do not contaminate water, food or feed by storage or disposal.

14 TRANSPORT INFORMATION

UN No.	3017
Class	6.1
Packing group	III
Marine Pollutant	Yes
Shipping Description	Organophosphorus Pesticide Liquid, Toxic, N.O.S, Flammable

15 REGULATORY INFORMATION

Classification	Xn (Harmful) , N (Dangerous for the environment)
Risk Phrases	R20/21/22 - Harmful by inhalation, in contact and if swallowed R50/53- Toxic to aquatic organism and May cause long term adverse effects in the aquatic organism
Safety Phrases	S2 - Keep out of reach of children S24 - Avoid contact with skin S60 - This material and its container must be disposed as hazardous waste S61 Avoid release to the environment. Refer to special instructions / safety data sheets.

16 OTHER INFORMATION

MSDS First issued	14 th October 2009
MSDS revised	10 th February 2017

Disclaimer

The above information is intended to give general health and safety guidance on the storage and transport of the substance or product to which it relates. The requirement or recommendation of any relevant site or working procedure, system or policy in force or arising from any risk assessment involving the substance or product should take precedence over any of the guidance contained in this safety data sheet where there is a difference in the information given. The information provided in this safety data sheet is accurate at the date of publication, and will be updated as and when appropriate. No liability will be accepted for any injury loss or damage resulting from any failure to take account of information or advice contained in this safety data sheet
